Indiana University
Fall 2021

SPEA-E 552
A. Ward

Name: SOLUTION + GRADE NOTES

Quiz #2 (Lectures 7-10) (28 POINTS POSSIBLE)

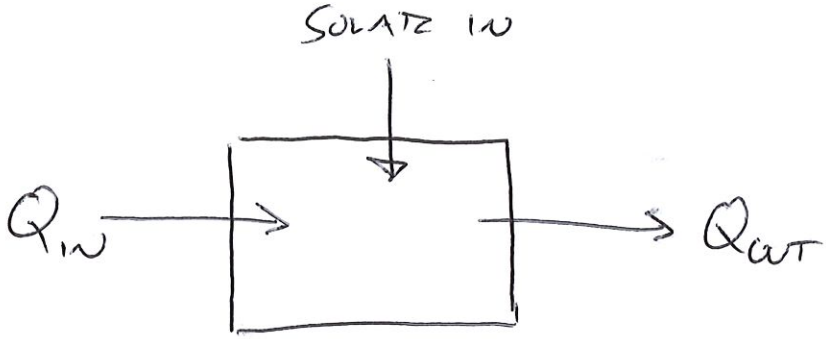
1. What is the difference between the Maximum Contaminant Level (MCL) and Maximum Contaminant Level Goal (MCLG) in USEPA drinking water standards? Why aren't these the same?

MCL = MAX. ALLOWABLE IN DRINKING WATER +1

MCLG : MAX. TO HAVE NO HEALTH IMPACT IN DRINKING WATER +1

MCL ≠ MCLG BECAUSE IT MAY NOT BE FEASIBLE (NOT POSSIBLE, TOO COSTLY) TO ACHIEVE MCLG. +2 FOR EXPLANATION

2. A 2-m x 2-m solar collector is used to heat water flowing into the device at 10 deg. C at a rate of 0.2 m^3/s. The solar collector receives 500 W/m^2 of incident solar radiation, of which 50% is transferred to the water. What is the temperature of the water flowing out of the collector?



+1 EFFORT
+1 E BALANCE
+1 $Q_{IN} \rightarrow E_{IN}$
+1 SOLAR $\rightarrow E_{IN}$
+1 $Q_{OUT} \rightarrow E_{OUT}$
+1 $\dot{m} = Q\rho$
+1 CALC
+1 UNITS

$\Sigma = 8$ POINTS

$$E_{IN} = E_{OUT}$$

$$\downarrow$$

$$Q\rho C_p T_{IN} + E_{SOLAR} = Q\rho C_p T_{OUT}$$

$$\downarrow$$

$$0.2\,\frac{m^3}{s} \cdot 999\,kg/m^3 \cdot 4186\,\frac{J}{kg \cdot {}^\circ C} * 10^\circ C + 0.50 * \frac{500\,W}{m^2} \cdot 2m \cdot 2m$$

$$= 0.2\,\frac{m^3}{s} \cdot 999\,kg/m^3 \cdot 4186\,\frac{J}{kg \cdot {}^\circ C} \cdot T_{OUT}$$

$\downarrow$ CHECK: EACH TERM IS J/s $\left(1\,W = 1\,\frac{J}{s}\right)$

SOLVE TO $\boxed{T_{OUT} = 10.0012\,{}^\circ C}$

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3. You are reviewing the design for a set of four rapid mix basins operating in parallel (i.e., ¼ of discharge through each basin) will treat a total of 20,000 m^3/d. The water has a viscosity of 1.2×10^{-3} kg/(m*s).

A. For a desired contact time of 20-s in the rapid mix basin, what is the volume required for each of the four basins?

+1 EQ
+1 CALC
+1 UNITS

$$\tau = \frac{\forall}{Q} \rightarrow 20s = \frac{\forall}{\frac{1}{4}\cdot 20{,}000\,\frac{m^3}{d}\cdot\frac{1\,d}{86400\,s}} \rightarrow \boxed{\forall = 1.16\,m^3}$$

B. The current design calls for each basin to be mixed by a 10 hp motor (i.e., the power input to the motor is 10 hp). What will be the velocity gradient (G) in the basin? Report your answer in units of s^{-1}.

+1 EQ
+1 P UNITS
+1 CALC
+1 UNITS

UNITS: $10\,hp \rightarrow 7457\,W = 7457\,\frac{N\cdot m}{s} = 7457\,\frac{kg\cdot m^2}{s^2}$ } UNIT CONVERSIONS

$$G = \sqrt{\frac{P}{\mu\forall}} = \sqrt{\frac{7454\,kg\cdot m^2/s^2}{1.2\times10^{-3}\,\frac{kg}{m\cdot s}\times 1.16\,m^3}} = \boxed{2315\,s^{-1}}$$

C. Typical design parameters for a basin would specify two conditions are met:
$600 < G < 1000$
$5*10^4 < G\tau < 5*10^5$
Does the proposed design meet these criteria?

+2 G CHECK

$\boxed{\text{G NOT MET}}$ (per part B)

+2 Gτ CHECK

$G\tau = 2315/s \cdot 20s = 46{,}300 = 4.63\times10^4$ $\boxed{G\tau \text{ NOT MET}}$

D. Based on your responses above, would you recommend a larger or smaller motor, and why? What would change, if anything, if you take efficiency into account?

+2 P IS TOO LARGE FOR G AS CALCULATED. USE A SMALLER MOTOR.

+3 WE ASSUMED 100% EFFICIENCY. IF MOTOR ISN'T 100% EFFICIENT, P↓ => G↓ => Gτ↓. SMALLER Gτ STILL WON'T MET CRITERIA, SO THIS WON'T FIX ALL PROBLEMS.